

# Jiannan LI

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## Current position

Assistant Professor

School of Computing and Information Systems, Singapore Management University

## Research Interest

Human-AI Interaction, Human-Robot Interaction

## Education

|           |                                                             |
|-----------|-------------------------------------------------------------|
| 2016-2023 | PhD in Computer Science, University of Toronto              |
| 2012-15   | MSc in Computer Science, University of Calgary              |
| 2008-12   | BENG in Automation and Control, Southeast University, China |

## Publications

### PEER-REVIEWED CONFERENCE AND JOURNAL ARTICLES

- DIS[C.21] Tianyi Zhang, Emran Poh, Yueyue Hou, Yi-Chieh Lee, Renwen Zhang, Jiannan Li, Anthony Tang. 2026. **“Grandpa, Can You Speak Nicer?”: Envisioned Chatbot Roles and Design Tensions in Intergenerational Communication Conflicts.** In *Proceedings of the 2026 Designing Interactive Systems Conference*.
- DIS[C.20] ShunYi Yeo, Tianyi Zhang, Scott Bateman, Gary Hsieh, Young-Ho Kim, Simon Tangi Perrault, Jiannan Li, Anthony Tang. 2026. **Group Conversational Agents: A Review of Designs that Support and Shape Group Interaction.** In *Proceedings of the 2026 Designing Interactive Systems Conference*. 🏆 Honorable Mention Award (top 5%).
- TVCG[J.7] Shaolun Ruan, Feng Liang, Rohan Ramakrishna, Chao Ren, Rudai Yan, Qiang Guan, Jiannan Li, Yong Wang. 2026. **Towards Explainable Quantum AI: Informing the Encoder Selection of Quantum Neural Networks via Visualization.** In *IEEE Transactions on Visualization and Computer Graphics*.
- SenSys[C.19] Emran Poh, Yueyue Hou, Tianyi Zhang, Jiannan Li. 2026. **From Cheap to Chic: Enhancing Music Playback Quality of Budget Earphones via Hardware-Aware Learning.** In *Proceedings of the 2026 ACM/IEEE International Conference on Embedded Artificial Intelligence and Sensing Systems*.
- CHI[C.18] Emran Poh, Yueyue Hou, Tianyi Zhang, Jiannan Li. 2026. **‘Show It, Don’t Just Say It’: The Complementary Effects of Instruction Multimodality for Software Guidance.** In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*.

- ICHMS[C.17] Ruochen Li, Ziyi Chang, Junyan Hu, Jiannan Li, Amir Atapour-Abarghouei, Hubert PH Shum. 2026. **ART: Adaptive Relational Transformer for Pedestrian Trajectory Prediction with Temporal-Aware Relations**. In *The 6th IEEE International Conference on Human-Machine Systems*.
- TVCG[J.6] Manusha Karunathilaka, Litian Lei, Yiming Gao, Yong Wang, Jiannan Li. 2026. **Compendia: Automated Visual Storytelling Generation from Online Article Collection**. In *IEEE Transactions on Visualization and Computer Graphics*.
- TVCG[J.5] Shaolun Ruan, Rui Sheng, Xiaolin Wen, Jiachen Wang, Tianyi Zhang, Yong Wang, Tim Dwyer, Jiannan Li. 2025. **Qualitative Study for LLM-assisted Design Study Process: Strategies, Challenges, and Roles**. In *IEEE Transactions on Visualization and Computer Graphics*. 🏆 Honorable Mention Award (top 5%) at VIS 25.
- IMWUT[J.4] Ruyi Li, Jingfei Guo, Xinyi Zhang, Xuji Zhang, Zeqing Li, Jiannan Li, Jiangtao Gong. 2025. **Guiding Multiple Remote Users in Physical Tasks with Language-driven Robotic Telepresence**. In *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*.
- CSCW[J.3] Qian Wan, Jiannan Li, Huanchen Wang, Zhicong Lu. 2025. **Polymind: Parallel Visual Diagramming with Large Language Models to Support Prewriting Through Microtasks**. In *Proceedings of the ACM on Human-Computer Interaction, CSCW '25*.
- CHI[C.16] Tianyi Zhang, Colin Au Yeung, Emily Aurelia, Yuki Onishi, Neil Chulpongsatorn, Jiannan Li, Anthony Tang. 2025. **Prompting an Embodied AI Agent: How Embodiment and Multi-modal Signaling Affects Prompting Behaviour**. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. 🏆 Honorable Mention Award (top 5%).
- HRI[C.15] Karthik Mahadevan, Blaine Lewis, Jiannan Li, Bilge Mutlu, Anthony Tang, Tovi Grossman. 2025. **ImageInThat: Manipulating Images to Convey User Instructions to Robots**. In *Proceedings of the 2025 IEEE/ACM International Conference on Human-Robot Interaction*. 🏆 Best Paper Award (top 1%).
- UIST[C.14] Ludwig Sidenmark, Tianyu Zhang, Leen Al Labadidi, Jiannan Li, Tovi Grossman. 2024. **Desk2Desk: Optimization-based Mixed Reality Workspace Integration for Remote Side-by-side Collaboration**. In *Proceedings of the 2024 Annual ACM Symposium on User Interface Software and Technology*. Acceptance rate 24%.
- DIS[C.13] Setareh Aghel Manesh, Tianyi Zhang, Yuki Onishi, Kotaro Hara, Scott Bateman, Jiannan Li, and Anthony Tang. 2024. **How People Prompt to Create Interactive VR Scenes**. In *Proceedings of the 2024 ACM Conference on Designing Interactive Systems*. Acceptance rate 27.4%.
- CHI[C.12] Jiannan Li, Maurício Sousa, Karthik Mahadevan, Bryan Wang, Paula Akemi Aoyau, Nicole Yu, Angela Yang, Ravin Balakrishnan, Anthony Tang, and Tovi Grossman. 2023. **Stargazer: An Interactive Camera Robot for Capturing How-To Videos Based on Subtle Instructor Cues**. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. Acceptance rate 28.4%.
- VR[C.11] Sixuan Wu, Jiannan Li, Mauricio Sousa, and Tovi Grossman. 2023. **Investigating Guardian Awareness Techniques to Promote Safety in Virtual Reality**. In *Proceedings of the 30th IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR '23)*.
- CSCW[J.2] Kartikaeya Kumar, Lev Poretski, Jiannan Li, and Anthony Tang. 2022. **Tourgether360: Collaborative Exploration of 360° Videos Using Pseudo-Spatial Navigation**. In *Proceedings of the ACM on Human Computer Interaction*.
- CHI[C.10] Kevin Huang, Jiannan Li, Maurício Sousa, and Tovi Grossman. 2022. **immersivePOV: Filming How-To Videos with a Head-Mounted 360° Action Camera**. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 🏆 Honorable Mention Award (top 5%).

- CHI[C.9] Jiannan Li, Maurício Sousa, Chu Li, Jessie Liu, Yan Chen, Ravin Balakrishnan, and Tovi Grossman. 2022. **ASTEROIDS: Exploring Swarms of Mini-Telepresence Robots for Physical Skill Demonstration**. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* Accepted with minor revision, acceptance rate 12.5%.
- UIST[C.8] Jiannan Li, Jiahe Lyu, Maurício Sousa, Ravin Balakrishnan, Anthony Tang, and Tovi Grossman. 2021. **Route Tapestries: Route Tapestries: Navigating 360 Virtual Tour Videos Using Slit-Scan Visualizations**. In *Proceedings of the 34th Annual ACM Symposium on User Interface Software and Technology*. Acceptance rate 25.9%.
- UIST[C.7] Jiangtao Gong, Teng Han, Siling Guo, Jiannan Li, Siyu Zha. 2021. **HoloBoard: an Immersive Teaching Board for Lecture-based Classes**. In *Proceedings of the 34th Annual ACM Symposium on User Interface Software and Technology*. Acceptance rate 25.9%.
- CHI[C.6] Zhicong Lu, Chenxinran Shen, Jiannan Li, Hong Shen, and Daniel Wigdor. 2021. **More Kawaii than a Real-Person Live Streamer: Understanding How the Otaku Community Engages with and Perceives Virtual YouTubers**. In *Proceedings of 2021 SIGCHI Conference on Human Factors in Computing Systems*. Acceptance rate 26.3%.
- GI[C.5] Jiannan Li, Ravin Balakrishnan, and Tovi Grossman. 2020. **StarHopper: A Touch Interface for Remote Object-Centric Drone Navigation**. In *Proceedings of the 2020 Graphical Interface Conference*.
- CHI[C.4] Teng Han, Jie Liu, Khalad Hasan, Mingming Fan, Junhyeok Kim, Jiannan Li, Xiangmin Fan, Feng Tian, Edward Lank, and Pourang Irani. 2019. **PinchList: Leveraging Pinch Gestures for Hierarchical List Navigation on Smartphones**. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. Acceptance rate 23.8%.
- CHI[C.3] Teng Han, Jiannan Li, Khalad Hasan, Keisuke Nakamura, Randy Gomez, Ravin Balakrishnan, and Pourang Irani. 2018. **PageFlip: Leveraging Page-Flipping Gestures for Efficient Command and Value Selection on Smartwatches**. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. Acceptance rate 25.7%.
- IJHCS[J.1] Jiannan Li, Saul Greenberg, Ehud Sharlin. 2017. **A Two-Sided Collaborative Transparent Display Supporting Workspace Awareness**. In *International Journal of Human Computer Studies*, 101:23-44.
- DIS[C.2] Jiannan Li, Saul Greenberg, Ehud Sharlin, and Joaquim Jorge. 2014. **Interactive two-sided transparent displays: designing for collaboration**. In *Proceedings of the 2014 conference on Designing interactive systems*. Acceptance rate 27%.
- ICCSNT[C.1] Jiannan Li, Pengbo Li, Hongyi Liu, Dong Li, and Jian Tang. 2011. **A Contactless Battery Charging and Monitoring System for Wireless Sensor Network Nodes**. In *Proceedings of 2011 International Conference on Computer Science and Network Technology*.

#### PRE-PRINTS

- arXiv Rahul Arora, Jiannan Li, Gongyi Shi, and Karan Singh. 2021. **Thinking Outside the Lab: VR Size & Depth Perception in the Wild**. arXiv:2105.00584.

#### POSTERS & WORKSHOPS

- CHILBW Nikhita Joshi, Richard Li, Jiannan Li, Leonardo Pavanatto, Michel Pahud, Jatin Sharma, Bongshin Lee, Hugo Romat, William A.S. Buxton, Nicolai Marquardt, Ken Hinckley, Nathalie Riche.

- Opportunistic Nudges for Task Migration Between Personal Devices.** In *CHI '24 Extended Abstracts on Human Factors in Computing Systems*
- AIVR Zhuoyue Lyu, Jiannan Li, Bryan Wang. 2021. **ALive: Interactive Visualization and Sonification of Neural Networks in Virtual Reality.** In *2021 IEEE International Conference on Artificial Intelligence and Virtual Reality (AIVR).*
- HAI Jiannan Li, Maurício Sousa, Ravin Balakrishnan, and Tovi Grossman. 2021. **Constellation: a Multi-User Interface for Remote Drone Tours.** In *Proceedings of the 9th International Conference on Human-Agent Interaction (HAI '21).*
- CHIEA Jiannan Li, Ehud Sharlin, Saul Greenberg, and Michael Rounding. 2013. **Designing the car iWindow: exploring interaction through vehicle side windows.** In *CHI '13 Extended Abstracts on Human Factors in Computing Systems.*

## Research Internships

- 2022 Microsoft Research, Redmond, US  
With Ken Hinckley and Nathalie Henry Riche in the EPIC reserach group.
- 2018 Université Paris-Saclay & INRIA, Orsay, France  
With Cédric Fleury and Michel Beaudouin-Lafon, on sense-making on wall-sized displays.
- 2014 Lenovo Research, Beijing, China  
With Xiang Cao, on novel interaction on mobile phone edges.

## Industry Employment

- 2015-16 Software engineer at Nureva Inc., Calgary, Canada

## Awards

- 2022 Snap Creative Challenge Award
- 2021 Wolfond Scholarship in Wireless Information Technology (5,000 CAD)
- 2016-20 Ontario Trillium Scholarship (40,000 CAD per year)
- 2016-18 Wolfond Scholarship in Wireless Information Technology (20,000 CAD)
- 2013,14 University of Calgary Computer Science Graduate Research Award (2,000 CAD)

## Services

### COMMITTEE MEMBER

- 2025,26 Technical Program Committee member, ACM UIST
- 2024-26 Technical Program Committee member, ACM CHI
- 2022-24 Technical Program Committee member, ACM ICMI
- 2026 Interactivity Chair, ACM DIS
- 2021 Web Chair, IEEE VR

## REVIEWER

|            |                                                                            |
|------------|----------------------------------------------------------------------------|
| 2017,20-24 | ACM CHI, special recognition for outstanding reviews for CHI '22           |
| 2020,22-24 | ACM UIST, special recognition for outstanding reviews for UIST '22 (2)     |
| 2020,22-24 | ACM CSCW, special recognition for outstanding reviews for CSCW '23 and '24 |
| 2023,24    | ACM IMWUT                                                                  |
| 2021,22,24 | IEEE VR                                                                    |
| 2024       | IEEE ISMAR                                                                 |
| 2023,24    | ACM DIS                                                                    |
| 2024       | ACM CHI PLAY                                                               |
| 2022       | ACM VRST                                                                   |
| 2014,15,22 | ACM TEI                                                                    |
| 2016,17    | IEEE RO-MAN                                                                |
| 2025       | ACM Transactions on Human-Robot Interaction                                |
| 2023       | ACM Transactions on Computer-Human Interaction                             |
| 2021       | IEEE Transactions on Human-Machine Systems                                 |
| 2020       | International Journal on Human-Computer Studies                            |

## Supervision & Mentoring

### PHD STUDENT

|           |                                                        |
|-----------|--------------------------------------------------------|
| 2023-     | Tianyi Zhang (SMU, co-advised with Tony Tang)          |
| 2024-     | Yueyue Hou (SMU)                                       |
| 2024-     | Manusha Karunathilaka (SMU, co-advised with Wang Yong) |
| 2024-2026 | Shaolun Ruan (SMU, co-advised with Wang Yong)          |
| 2025-     | Emran Poh (SMU)                                        |

### MASTER STUDENT

|      |                                                                    |
|------|--------------------------------------------------------------------|
| 2023 | Setareh Aghel Manesh (SMU, visiting student from UCalgary), [C.13] |
| 2022 | Paula Akemi Aoyagui (UToronto), [C.12]                             |
| 2022 | Nicole Sin Ting Yu (UToronto), [C.12]                              |
| 2021 | Chu Li (UToronto), [C.9]                                           |

### UNDERGRADUATE STUDENT

|         |                                        |
|---------|----------------------------------------|
| 2022    | Angela Yang (UToronto), [C.12]         |
| 2021,22 | Sixuan Wu (UToronto), [C.11]           |
| 2021,22 | Yuchen Fan (UToronto)                  |
| 2021    | Kartikaeya Kumar (IIT Guwahati), [J.2] |
| 2021    | Kevin Huang (UToronto), [C.10]         |
| 2020,21 | Jiahe Lyu (UToronto), [C.8]            |

## Teaching

2024-26    IS133 (Web Application Development)  
2025       CS616 (Human-AI Interaction Design)

## References

Available upon request.